Anti-derivatives of exponential funct

1 Find the primitive function of the following:

a $f'(x) = e^{2x}$

b $f'(x) = e^{-5x}$

c $f'(x) = e^{0.25x}$

d $f'(x) = 9e^{3x}$

e $f'(x) = e^{x} + e^{-3x}$

f $f'(x) = e^{3x} - e^{-8x}$

g $f'(x) = \frac{1}{6}\left(e^{0.5x} + e^{3x}\right)$

h $f'(x) = e^{2x} + \sqrt{x}$

2 Find the following indefinite integrals:

a $\int e^{-x} dx$

b $\int e^{2x} dx$

c $\int e^{\frac{1}{2}x} dx$

d $\int 4e^{2x} dx$

e $\int e^{4-3x} dx$

f $\int e^{3-2x} dx$

g $\int 4e^{1-2x} dx$

h $\int \left(e^{0.5x} + e^{3x}\right) dx$

i $\int \left(e^{t} - 5\right) dt$

j $\int \left(e^{4t} + t^{2}\right) dt$

k $\int \left(e^{3v} + v^{4}\right) dv$

3 Consider the function $y = x^2 e^x$.

a Find an expression for $\frac{dy}{dx}$.

b Hence find $\int x\left(2+x\right) e^{x} dx$.

Find value of C

4 A curve has gradient function $\frac{dy}{dx} = e^{3x}$. Find the equation of the curve if it passes through the point $(0, -1)$.

5 Consider gradient function $\frac{dy}{dx} = 8e^{4x} - 5$.

a Find $\int (8e^{4x} - 5) dx$.

b If $y = 7$ when $x = 0$, find y in terms of x.

6 Consider $f'\left(x\right) = e^{-2x} + 6\sqrt{x}$.

a Find $\int \left(e^{-2x} + 6\sqrt{x}\right) dx$.

b If $f\left(1\right) = 3$, find $f\left(x\right)$.

7 Consider $f'(x) = \dfrac{3e^{2x}+1}{e^x}$.

a Find $\int \left(\dfrac{3e^{2x}+1}{e^x}\right) dx$.

b If $f(0) = 4$, find $f(x)$.

8 Find the equation of the curve given the gradient function and a point on the curve:

a $\dfrac{dy}{dx} = e^{2x}$, and point $(0, 6)$.

b $\dfrac{dy}{dx} = e^x - 2x$, and point $(0, 4)$.

9 A curve has gradient function $\dfrac{dy}{dx} = e^{kx}$ for some constant k. The point $\left(1, e^3\right)$ lies both on the gradient function and also the original curve y.

a Determine the value of k.

b Find y in terms of x.

10 Find y in terms of x given the following information:

- $\dfrac{dy}{dx} = ke^x + 2$, for some constant k
- When $x = 0 :\ y = -1$, and $\dfrac{dy}{dx} = 5$